



PERSONALISED E-PAPER CLOCK

Using ESP32 Microcontroller

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E-paper is continuously evolving. The day is not far when e-papers will replace the current OLED screens of mobiles and other handheld devices, resulting in significant reductions in power consumption and increased usage time.

We are fascinated by the vivid colours of the e-paper, which remain displayed even when the power supply is removed, true for both black-and-white as well as coloured e-papers. The > 170-degree viewing angle is an extra advantage, and with no backlight, e-papers reflect the ambient light incident on them, resulting in extremely low power consumption.

However, one significant drawback of the e-paper is its slow refresh rate, ranging from 3 to 8 seconds. This is the main reason why e-papers have not yet entered the mobile screen arena; they are mostly limited to e-readers and clock displays, which feature only hour and minute displays. Fig. 1 displays the author's prototype. The components (bill of materials) used in this clock are listed below.

BILL OF MATERIALS

Items	Quantity	Description
ESP32 – MCU (MOD1)	1	Microcontroller
3.3V voltage regulator (IC1)	1	HT7333A regulator
19cm (7.5-inch) black-and-white e-paper (MOD2)	1	Display
Wires, PCB, connector		
10 μ F, 16V electrolytic capacitors (C1, C2)	2	
RTC module (MOD3)	1	DS3231

Circuit and working

Fig. 2 illustrates the circuit diagram of the clock. The clock is built around ESP32 MCU (MOD1), 19cm black-and-white e-paper display (MOD2), DS3231 RTC module (MOD3), 3.3V regulator HT7333A (IC1), and a few other components.

This clock primarily employs a 19cm black-and-white e-paper (800 × 480 pixel area), featuring an

analogue clock view with seconds display. The seconds hand does not move every second but in every two seconds, and sometimes in three seconds. The entire circuit draws about 60mA current only from the 3.7V LiPo cell and so runs continuously for over two days. In the event of a power failure, the one-cell backup is more than sufficient!

Software coding

For coding, the required library files are to be installed in the library's directory of Arduino. The files can be found along with the code on electronicsforu.com. If a different e-paper display is used than the one suggested here, just uncomment and use the right display driver. The RTC clock is used to keep track of time and date, and in the code, the I2C address is set accordingly. The default I2C address for most RTCs is 0X68.

Unlike normal Arduino programming where everything inside the loop {} repeats, in e-paper programming not everything can be repeated to make it work. e-paper has a display divided into two pages: the first page and the next page. You have to write on these pages in reverse colour so that when the first page expires and the next page starts, the

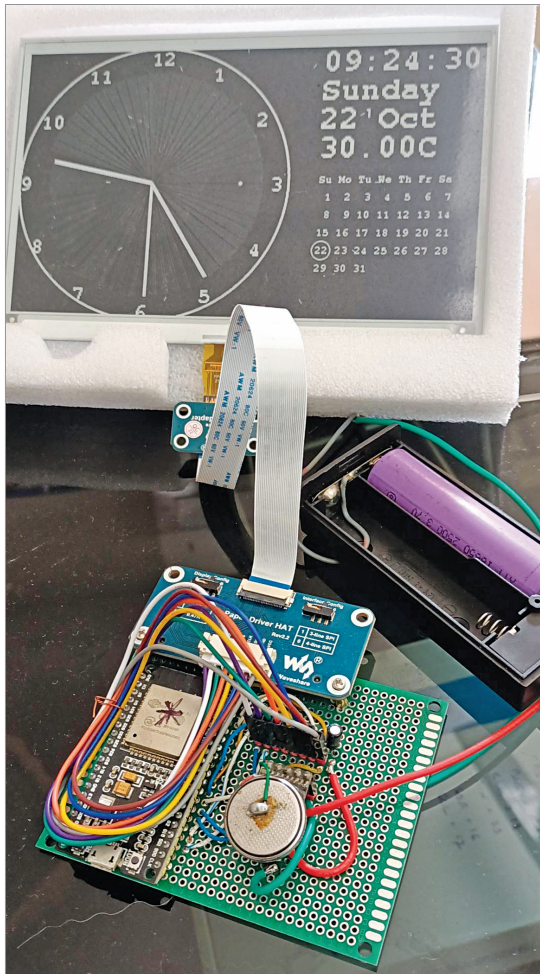


Fig. 1: Author's prototype